$[Xe]\ 4f^0\ 5d^1\ 6s^2$ and it belongs to d block. Filling of 4f orbital starts from Cerium (Ce) and its electronic configuration is $[Xe]\ 4f^1\ 5d^1\ 6s^2$. As we move from Cerium to other elements the additional electrons are progressively filled in 4f orbitals as shown in the table.

Table : electronic configuration of Lanthanum and Lanthanoids

Name of the element	Atomic number	Symbol	Electronic configuration
Lanthanum	57	La	$[Xe]\ 4f^0\ 5d^1\ 6s^2$
Cerium	58	Ce	$[Xe]\ 4f^1\ 5d^1\ 6s^2$
Praseodymium	59	Pr	$[Xe]\ 4f^3\ 5d^0\ 6s^2$
Neodymium	60	Nd	$[Xe]\ 4f^4\ 5d^0\ 6s^2$
Promethium	61	Pm	$[Xe]\ 4f^5\ 5d^0\ 6s^2$
Samarium	62	Sm	$[Xe]\ 4f^6\ 5d^0\ 6s^2$
Europium	63	Eu	$[Xe]\ 4f^7\ 5d^0\ 6s^2$
Gadolinium	64	Gd	$[Xe]\ 4f^7\ 5d^1\ 6s^2$
Terbium	65	Tb	$[Xe]\ 4f^9\ 5d^0\ 6s^2$
Dysprosium	66	Dy	$[Xe]\ 4f^{10}\ 5d^0\ 6s^2$
Holmium	67	Ho	$[Xe]\ 4f^{11}\ 5d^0\ 6s^2$
Erbium	68	Er	$[Xe]\ 4f^{12}\ 5d^0\ 6s^2$
Thulium	69	Tm	$[Xe]\ 4f^{13}\ 5d^0\ 6s^2$
Ytterbium	70	Yb	$[Xe]\ 4f^{14}\ 5d^0\ 6s^2$
Lutetium	71	Lu	$[Xe]\ 4f^{14}\ 5d^1\ 6s^2$

In Gadolinium (Gd) and Lutetium (Lu) the 4f orbitals, are half-filled and completely filled, and one electron enters 5d orbitals. Hence the general electronic configuration of 4f series of elements can be written as $[Xe]\ 4f^{1-14}\ 5d^{0-1}\ 6s^2$

Oxidation state of lanthanoids:

The common oxidation state of lanthanoids is +3. In addition to that some of the lanthanoids also show either +2 or +4 oxidation states.

Gd^{3+} and Lu^{3+} ions have extra stability, it is due to the fact that they have exactly half filled and completely filled f-orbitals respectively. Their electronic configurations are

$$Gd^{3+} : [Xe]4f^7$$
$$Lu^{3+} : [Xe]4f^{14}$$

XII U4-D-Block-Jerald Folder.indd 120 22-12-2022 19:49:34